Lecture 6

Characters

Objectives. *Lecture 5 counted the copies of an irreducible U in V by $\dim \mathrm{Hom}_G(U,V)$. This lecture shows that the traces of the operators in a representation determine it up to isomorphism. Averaging on $\mathrm{Hom}(U,V)$ gives $\dim \mathrm{Hom}_G(U,V) = \langle \chi_U, \chi_V \rangle_G$, for the characters $\chi(g) = \operatorname{tr} \rho(g)$, functions on the classes; so the irreducible characters are orthonormal. With the characters of duals, tensor products and quotients, we build the table of S_4.*

By the end of this lecture you will be able to:

- Compute characters, and decide from their inner products whether a representation is irreducible and how it decomposes.
- Prove that $\dim \mathrm{Hom}_G(U,V) = \langle \chi_U, \chi_V \rangle_G$ by averaging on $\mathrm{Hom}(U,V)$, and deduce that the irreducible characters are orthonormal and that a representation is determined up to isomorphism by its character.
- Compute the characters of V^*, $V \otimes W$, $\mathrm{Sym}^2(V)$, $\Lambda^2(V)$ and of a lift from a quotient, and prove that $|\chi(g)| \leq \chi(e)$, with equality exactly when $\rho(g)$ is a scalar.
- Build the character table of S_4 from the trivial, sign and permutation representations and a lift from $S_4/V_4 \cong S_3$, and prove that it lists every irreducible character.

6.1 Characters and class functions

Throughout, G is finite, and representations are complex and finite-dimensional.

Definition 6.1 (Characters). The *character* of a representation (ρ, V) is the function

$$\chi_V : G \to \mathbb{C}, \qquad \chi_V(g) = \operatorname{tr} \rho(g).$$

Its *degree* is $\chi_V(e) = \dim V$. It is *irreducible* if V is, and *linear* if $\dim V = 1$; then it is the homomorphism $\rho : G \to \mathbb{C}^\times$ itself.

The trivial representation has the constant character 1_G.

Proposition 6.2 (First properties). *Let V and W be representations of G, and $g, h \in G$.*

(i) $\chi_V(hgh^{-1}) = \chi_V(g)$.

(ii) $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

(iii) If $V \cong W$, then $\chi_V = \chi_W$. On $V \oplus W$, with $g \cdot (v,w) = (g \cdot v, g \cdot w)$, $\chi_{V \oplus W} = \chi_V + \chi_W$.

Proof. (i) $\rho(hgh^{-1}) = \rho(h)\rho(g)\rho(h)^{-1}$, and conjugate matrices have the same trace.

(ii) By Corollary 5.17(ii), $\rho(g)$ is diagonalisable, with eigenvalues $c_1, \ldots, c_d$ that are roots

of unity; so $|c_i| = 1$ and $c_i^{-1} = \overline{c_i}$. The eigenvalues of $\rho(g^{-1}) = \rho(g)^{-1}$ are the c_i^{-1}, so

$$\chi_V(g^{-1}) = \sum_i c_i^{-1} = \sum_i \overline{c_i} = \overline{\chi_V(g)}.$$

(iii) Isomorphic representations differ by a change of basis (§ 5.1), which keeps traces. In a basis of V followed by one of W, the matrix of g on $V \oplus W$ is block diagonal, with blocks $D_V(g)$ and $D_W(g)$. ∎

Definition 6.3 (Class functions)**.** A *class function* is a function $\varphi : G \to \mathbb{C}$ that is constant on each conjugacy class. The class functions form a vector space $\mathrm{CF}(G)$, with the inner product

$$\langle \varphi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\varphi(g)}\, \psi(g).$$

The indicator functions of the classes are a basis of $\mathrm{CF}(G)$, so $\dim \mathrm{CF}(G)$ is the number of classes.

The form is antilinear in the first slot and linear in the second, and $\langle \varphi, \varphi \rangle_G > 0$ for $\varphi \neq 0$.[1] We call $\langle \varphi, \varphi \rangle_G$ the *squared norm* of φ; "norm one" means squared norm 1. Characters are class functions by conjugacy invariance (Proposition 6.2). Summing over classes, with g a representative of each class,

$$\langle \varphi, \psi \rangle_G = \frac{1}{|G|} \sum_{\text{classes } [g]} |[g]|\, \overline{\varphi(g)}\, \psi(g).$$

Example 6.4 (The characters of S_3)**.** We compute the characters of the trivial representation, the sign and the plane E on the three classes of S_3 in Example 1.21. Recall from Lecture 5 that $E \subseteq \mathbb{C}^3$ is the plane of coordinate sum zero, with basis $f_1 = e_1 - e_2$, $f_2 = e_2 - e_3$. Write $\rho_E : S_3 \to \mathrm{GL}(E)$ for the action on E. Taking traces of the matrices in § 5.5 gives:

class of	e	$(1\,2)$	$(1\,2\,3)$
size	1	3	2
1_G	1	1	1
sgn	1	-1	1
χ_E	2	0	-1

For instance, the diagonal entries of the matrix of $(1\,2)$ are -1 and 1, so its character value is 0. For $(1\,2\,3)$ they are 0 and -1, giving the value -1.

Example 6.5 (Guard: traces, not eigenvalues)**.** A character sums eigenvalues; in degree above 1 it need not be a homomorphism. $D_E((1\,2\,3))$ has characteristic polynomial $t^2 + t + 1$, with roots ζ, ζ^2 for $\zeta = \exp(2\pi \mathrm{i}/3)$, so

$$\chi_E\big((1\,2\,3)\big) = \zeta + \zeta^2 = -1.$$

Its square $(1\,3\,2)$ has the eigenvalues ζ^2, ζ^4, and

$$\chi_E\big((1\,2\,3)^2\big) = -1, \qquad \chi_E\big((1\,2\,3)\big)^2 = 1.$$

Thus a character need not be multiplicative. § 6.3 and Lecture 9 use $\chi(g^2)$.

[1] Serre and James–Liebeck put the conjugate on the second slot.

6.2 Averaging on Hom and orthonormality

For representations U and V, G acts on $\mathrm{Hom}(U,V)$ by $g \cdot T = \rho_V(g)\, T\, \rho_U(g)^{-1}$, with invariants $\mathrm{Hom}(U,V)^G = \mathrm{Hom}_G(U,V)$ (§ 5.2).

Lemma 6.6 (The character of Hom). *The character of* $\mathrm{Hom}(U,V)$ *is*

$$\chi_{\mathrm{Hom}(U,V)}(g) = \overline{\chi_U(g)}\,\chi_V(g).$$

Proof. In bases of U and V, $\mathrm{Hom}(U,V)$ is the space of $n \times m$ matrices, $n = \dim V$, $m = \dim U$, with basis the matrix units E_{kl} (1 in row k, column l, 0 elsewhere).

A trace. Let A be $n \times n$ and B be $m \times m$. The E_{kl}-coordinate of $AE_{kl}B$ is its (k,l) entry,

$$(AE_{kl}B)_{kl} = A_{kk}B_{ll},$$

and summing over k and l, the map $T \mapsto ATB$ has trace $\operatorname{tr} A \operatorname{tr} B$.

The character. Take $A = D_V(g)$ and $B = D_U(g^{-1})$. The trace of an inverse is the conjugate character value (Proposition 6.2), so

$$\chi_{\mathrm{Hom}(U,V)}(g) = \chi_V(g)\,\chi_U(g^{-1}) = \overline{\chi_U(g)}\,\chi_V(g). \qquad \blacksquare$$

Theorem 6.7 (Characters count intertwiners). *For representations U and V of G,*

$$\dim \mathrm{Hom}_G(U,V) = \langle \chi_U, \chi_V \rangle_G .$$

Proof. The dimension of the fixed subspace is the average of the traces of the representing operators (Proposition 5.8). Apply this to $\mathrm{Hom}(U,V)$ and use the character formula of Lemma 6.6:

$$\begin{aligned}
\dim \mathrm{Hom}_G(U,V) &= \dim \mathrm{Hom}(U,V)^G \\
&= \frac{1}{|G|} \sum_{g \in G} \chi_{\mathrm{Hom}(U,V)}(g) \\
&= \frac{1}{|G|} \sum_{g \in G} \overline{\chi_U(g)}\,\chi_V(g) \\
&= \langle \chi_U, \chi_V \rangle_G . \qquad \blacksquare
\end{aligned}$$

In particular, $\langle \chi_U, \chi_V \rangle_G$ is a non-negative integer.

Choose one representative of each isomorphism class of irreducible representations; write $\mathrm{Irr}(G)$ for this set.

Corollary 6.8 (Orthonormality). *Let $U, U' \in \mathrm{Irr}(G)$, and let V and W be representations of G.*

(i) $\langle \chi_U, \chi_{U'} \rangle_G$ *is* 1 *if* $U = U'$, *and* 0 *otherwise.*

(ii) $m_U(V) = \langle \chi_U, \chi_V \rangle_G$, *and* $\chi_V = \sum_U m_U(V)\,\chi_U$, *a finite sum over the U with $m_U(V) \neq 0$.*

(iii) $V \cong W$ *if and only if* $\chi_V = \chi_W$.

(iv) $\langle \chi_V, \chi_V \rangle_G = \sum_U m_U(V)^2$. *So V is irreducible if and only if* $\langle \chi_V, \chi_V \rangle_G = 1$.

(v) The irreducible characters are linearly independent in $\mathrm{CF}(G)$. *So* $\mathrm{Irr}(G)$ *has at most as many elements as G has classes; and if k irreducible representations have pairwise distinct characters, and k is the number of classes, they are all of* $\mathrm{Irr}(G)$, *up to isomorphism.*

Proof. (i) By Theorem 6.7 and Schur's lemma (Theorem 5.13(ii)), $\dim \mathrm{Hom}_G(U, U')$ is 1 or 0.

(ii) The first claim is Theorem 6.7 with Corollary 5.16. By Maschke's theorem, $V = U_1 \oplus \cdots \oplus U_k$ with each U_j irreducible, so $\chi_V = \sum_j \chi_{U_j}$ by Proposition 6.2(iii); and exactly $m_U(V)$ of the U_j are isomorphic to U.

(iii) One direction is Proposition 6.2(iii). Conversely, if $\chi_V = \chi_W$, the multiplicity formula gives the same multiplicities, and $V \cong W$ by Corollary 5.16.

(iv) Expand both characters using the multiplicity formula, then orthonormality. The multiplicities are real integers, so first-slot antilinearity does not change them:

$$\begin{aligned}\langle \chi_V, \chi_V \rangle_G &= \sum_{U,U'} m_U(V)\, m_{U'}(V)\, \langle \chi_U, \chi_{U'} \rangle_G \\ &= \sum_U m_U(V)^2.\end{aligned}$$

A sum of squares of non-negative integers is 1 exactly when one term is 1 and the others are 0. By the multiplicity theorem (Corollary 5.16), that is exactly when V has one irreducible summand, with multiplicity 1. An irreducible V has one such summand, and conversely.

(v) *Independence.* If $\sum_U a_U \chi_U = 0$, a finite sum, take the inner product with $\chi_{U'}$ in the first slot. Linearity in the second slot and orthonormality give $a_{U'} = 0$.

The count. So there are at most $\dim \mathrm{CF}(G)$ irreducible characters, hence at most that many irreducibles, since characters determine isomorphism classes.

Completeness. If irreducible representations $U_1, \ldots, U_k$ have pairwise distinct characters, they are pairwise non-isomorphic. Replacing each by its representative in $\mathrm{Irr}(G)$, orthogonality shows that the characters are orthonormal. If $k = \dim \mathrm{CF}(G)$, these form a basis. Any irreducible $U \in \mathrm{Irr}(G)$ distinct from these representatives has character orthogonal to the basis, hence zero. But $\chi_U(e) = \dim U \geq 1$. ∎

Definition 6.9 (Character table)**.** A *character table* lists the irreducible characters, one per isomorphism class, as rows and the conjugacy classes as columns. Each entry is the character value on that class.

Example 6.10 (S_3 again)**.** We compute the squared norm of the character of E from the class sizes and values in Example 6.4:

$$\langle \chi_E, \chi_E \rangle_{S_3} = \frac{1}{6}(4 + 3 \cdot 0 + 2 \cdot 1) = 1,$$

so E is irreducible by the character-norm criterion. The three irreducible characters 1_G, sgn, χ_E are distinct, and S_3 has three classes, so completeness gives $\mathrm{Irr}(S_3) = \{\mathbf{1}, \mathrm{sgn}, E\}$.

Example 6.11 (Guard: norm one is not enough)**.** On S_3, the class function $-\chi_E$ has squared norm $\langle -\chi_E, -\chi_E \rangle_{S_3} = \langle \chi_E, \chi_E \rangle_{S_3} = 1$. Consider also $\varphi = (1_G + \mathrm{sgn})/\sqrt{2}$. By orthonormality,

$$\langle \varphi, \varphi \rangle_{S_3} = \frac{1}{2}(1 + 0 + 0 + 1) = 1.$$

But a character takes the value $\dim V$, a non-negative integer, at e, and these take -2 and $\sqrt{2}$.

What does hold: let $\psi = \sum_U a_U \chi_U$ with integers a_U and $\langle \psi, \psi \rangle_G = 1$. By orthonormality,

$$\sum_U a_U^2 = 1,$$

so one a_U is ± 1 and the others are 0: $\psi = \pm \chi_U$. If also $\psi(e) > 0$, then $\psi = \chi_U$. Lecture 13 meets such an integral combination of norm one, and uses exactly this.

6.3 Tensor products, duals, Sym^2 and Λ^2

Duals and tensor products build representations from known ones. We will use their characters to compute the symmetric and exterior squares of E.

Definition 6.12 (Duals and tensor products). Let (ρ, V) and (ρ', W) be representations of G.

- The *dual representation* is $V^* = \mathrm{Hom}(V, \mathbb{C})$, with $(g \cdot f)(v) = f(g^{-1} \cdot v)$. Write $\rho^*(g)$ for this operator. In the dual basis, its matrix is $D_{V^*}(g) = D_V(g^{-1})^{\mathsf{T}}$.
- The *tensor product* is $V \otimes W$, with $g \cdot (v \otimes w) = (g \cdot v) \otimes (g \cdot w)$, so $(\rho \otimes \rho')(g) = \rho(g) \otimes \rho'(g)$.
- On $V \otimes V$, the *swap* is $\tau(v \otimes v') = v' \otimes v$. The *symmetric* and *exterior squares* are

$$\mathrm{Sym}^2(V) = \{x : \tau x = x\}, \qquad \Lambda^2(V) = \{x : \tau x = -x\}.$$

V^* is $\mathrm{Hom}(V, \mathbf{1})$ with Lecture 5's action, so it is a representation. $V \otimes W$ is one because $(A \otimes B)(A' \otimes B') = AA' \otimes BB'$.

Theorem 6.13 (Characters of constructions). *Let V and W be representations of G, and $d = \dim V$.*

(i) $\chi_{V^*} = \overline{\chi_V}$.

(ii) $\chi_{V \otimes W} = \chi_V \chi_W$.

(iii) $V \otimes V = \mathrm{Sym}^2(V) \oplus \Lambda^2(V)$, *a direct sum of subrepresentations of dimensions $d(d+1)/2$ and $d(d-1)/2$, and*

$$\chi_{\mathrm{Sym}^2(V)}(g) = \frac{1}{2}\big(\chi_V(g)^2 + \chi_V(g^2)\big),$$
$$\chi_{\Lambda^2(V)}(g) = \frac{1}{2}\big(\chi_V(g)^2 - \chi_V(g^2)\big).$$

Proof. (i) is Lemma 6.6 for $\mathrm{Hom}(V, \mathbf{1})$, as $\chi_{\mathbf{1}} = 1_G$.

Fix g. By Corollary 5.17(ii), V has a basis $v_1, \dots, v_d$ with $g \cdot v_i = c_i v_i$, and W a basis $w_1, \dots, w_{d'}$, where $d' = \dim W$, with $g \cdot w_j = b_j w_j$.

(ii) g multiplies the basis vectors $v_i \otimes w_j$ of $V \otimes W$ by $c_i b_j$, so

$$\chi_{V \otimes W}(g) = \sum_{i,j} c_i b_j = \chi_V(g)\, \chi_W(g).$$

(iii) *The splitting.* On pure tensors,

$$\tau\big((g \cdot v) \otimes (g \cdot v')\big) = (g \cdot v') \otimes (g \cdot v) = g \cdot \tau(v \otimes v').$$

Pure tensors span $V \otimes V$, so τ is an intertwiner. Thus $\mathrm{Sym}^2(V) = \ker(\tau - \mathbb{1})$ and $\Lambda^2(V) = \ker(\tau + \mathbb{1})$ are subrepresentations. As $\tau^2 = \mathbb{1}$, each x splits as

$$x = \frac{1}{2}(x + \tau x) + \frac{1}{2}(x - \tau x),$$

in $\mathrm{Sym}^2(V)$ and $\Lambda^2(V)$; only 0 satisfies $\tau x = x = -x$.

Bases. The vectors

$$v_i \otimes v_j + v_j \otimes v_i \quad (i \le j) \quad \text{in } \mathrm{Sym}^2(V),$$
$$v_i \otimes v_j - v_j \otimes v_i \quad (i < j) \quad \text{in } \Lambda^2(V)$$

span $V \otimes V$: adding and subtracting the two vectors for $i < j$ recovers $v_i \otimes v_j$ and $v_j \otimes v_i$ after division by 2, and the vector for $i = j$ is $2v_i \otimes v_i$. There are

$$\frac{1}{2}d(d+1) + \frac{1}{2}d(d-1) = d^2$$

of them, so together they form a basis. As the two squares are a direct sum, each family is a basis of its square. The element g multiplies each by $c_i c_j$.

The characters. The eigenvalues of $\rho(g^2) = \rho(g)^2$ are the c_i^2, so $\chi_V(g^2) = \sum_i c_i^2$, and

$$\chi_V(g)^2 = \Big(\sum_i c_i\Big)^2 = \chi_V(g^2) + 2\sum_{i<j} c_i c_j.$$

So $\chi_{\Lambda^2(V)}(g) = \sum_{i<j} c_i c_j$ is the second formula, and the first is $\chi_{\mathrm{Sym}^2(V)} = \chi_V^2 - \chi_{\Lambda^2(V)}$, by (ii) and Proposition 6.2(iii). ∎

Example 6.14 (The squares of E). The squares of e, $(1\,2)$, $(1\,2\,3)$ are e, e, $(1\,3\,2)$, so by Example 6.4:

class of g	e	$(1\,2)$	$(1\,2\,3)$
$\chi_E(g)^2$	4	0	1
$\chi_E(g^2)$	2	2	−1
$\chi_{\mathrm{Sym}^2(E)}(g)$	3	1	0
$\chi_{\Lambda^2(E)}(g)$	1	−1	1

The multiplicity formula (Corollary 6.8) applied to the symmetric square gives

$$m_{\mathbf{1}} = \frac{1}{6}(3 + 3\cdot 1 + 2\cdot 0) = 1,$$
$$m_{\mathrm{sgn}} = \frac{1}{6}(3 - 3\cdot 1 + 2\cdot 0) = 0,$$
$$m_E = \frac{1}{6}(2\cdot 3 + 3\cdot 0\cdot 1 + 2\cdot(-1)\cdot 0) = 1.$$

Thus $\mathrm{Sym}^2(E) \cong \mathbf{1} \oplus E$. The exterior-square character equals sgn, so $\Lambda^2(E) \cong \mathrm{sgn}$ by the character criterion (Corollary 6.8).

6.4 Lifting from quotients

A representation of a quotient gives a representation of the original group by composition with the quotient map. We determine which ones arise this way.

Theorem 6.15 (Lifting). *Let $N \trianglelefteq G$, with quotient map $q\colon G \to G/N$.*

(i) If $\bar\rho$ is a representation of G/N on U, then $\rho = \bar\rho \circ q$ is a representation of G on U with $N \le \ker\rho$, and $\chi_\rho(g) = \chi_{\bar\rho}(gN)$. Every representation ρ of G with $N \le \ker\rho$ is $\bar\rho \circ q$ for exactly one $\bar\rho$.

(ii) $\bar\rho \circ q$ is irreducible if and only if $\bar\rho$ is, and $\bar\rho \circ q \cong \bar\rho' \circ q$ if and only if $\bar\rho \cong \bar\rho'$. So, up to isomorphism, lifting is a bijection from $\mathrm{Irr}(G/N)$ onto the irreducible representations of G with N in their kernel.

(iii) The linear characters of G are the lifts of the linear characters of G/G'.

Proof. (i) ρ is a composite of homomorphisms, trivial on $N = \ker q$, and $\rho(g) = \bar{\rho}(gN)$. Conversely, if $N \leq \ker \rho$, then $\rho\colon G \to \mathrm{GL}(U)$ factors through G/N in exactly one way, by Theorem 2.5(i).

(ii) As q is onto, the operators $\rho(g)$ are exactly the operators $\bar{\rho}(gN)$. So the two have the same subrepresentations, and a linear map intertwines the lifts exactly when it intertwines $\bar{\rho}$ and $\bar{\rho}'$. Conversely, if an irreducible ρ has $N \leq \ker \rho$, factorisation gives $\rho = \bar{\rho} \circ q$, and the same subrepresentations make $\bar{\rho}$ irreducible. This proves the stated bijection.

(iii) A linear character is a homomorphism $G \to \mathbb{C}^\times$, to an abelian group, so it factors through G/G', by Proposition 4.13(iii). Conversely, a lift of a linear character is linear, by (i). ∎

6.5 The size of character values

The character value $\chi(g)$ is a sum of eigenvalues on the unit circle. We determine when its modulus is as large as possible and what χ tells us about the kernel.

Definition 6.16 (Kernel of a character)**.** For a character χ of G, its *kernel* is

$$\ker \chi = \{g \in G : \chi(g) = \chi(e)\}.$$

Proposition 6.17 (The size of character values)**.** *Let (ρ, V) be a representation of G with character χ, and $g \in G$.*

(i) $|\chi(g)| \leq \chi(e)$, with equality if and only if $\rho(g)$ is a scalar, $\rho(g) = c\mathbb{1}$.

(ii) $\chi(g) = \chi(e)$ if and only if $\rho(g) = \mathbb{1}$. So $\ker \chi = \ker \rho$, a normal subgroup.

Proof. If $V = 0$, both parts hold, as $\mathrm{End}(0) = \{0\}$ and $\chi = 0$; let $d = \chi(e) > 0$.

(i) By Corollary 5.17(ii), $\rho(g)$ is diagonalisable, with eigenvalues $c_1, \dots, c_d$ of modulus 1. By the triangle inequality,

$$|\chi(g)| = \left|\sum_i c_i\right| \leq \sum_i |c_i| = d.$$

Equality holds exactly when the c_i are positive multiples of one another; being of modulus 1, they are then all equal to one c. A diagonalisable map with the single eigenvalue c is $c\mathbb{1}$. Conversely, if $\rho(g) = c\mathbb{1}$, then $|c| = 1$ as c is an eigenvalue, so $|\mathrm{tr}(c\mathbb{1})| = d$.

(ii) If $\chi(g) = d$, then $\rho(g) = c\mathbb{1}$ by (i), and $dc = d$ gives $c = 1$. The converse is clear, and $\ker \rho$ is normal, as the kernel of a homomorphism (§ 2.1). ∎

Example 6.18 (Equality outside the kernel)**.** In Lecture 5's irreducible plane for D_4, the central r^2 acts by $\rho(r^2) = -\mathbb{1}$, so

$$\chi(r^2) = -2 :$$

equality in the character-value bound, with $c = -1$, but $r^2 \notin \ker \chi$ by Proposition 6.17, since $\chi(r^2) = -2 \neq 2 = \chi(e)$.

Lecture 9's proof of Burnside's $p^a q^b$ theorem on solvability uses both parts.

6.6 Worked example

We build the character table of S_4 from the representations and quotient maps found in earlier lectures.

The classes. The five cycle types of § 2.6 give five conjugacy classes of S_4. Five distinct irreducible characters therefore complete the table (Corollary 6.8(v)).

The linear characters. The commutator subgroup is $S_4' = A_4$ and $S_4/A_4 \cong C_2$ (§ 4.5), and C_2 has two linear characters (Example 5.18). By Theorem 6.15(iii), S_4 has two, 1_G and sgn.

The permutation representation. On $\mathbb{C}^4$, $\chi(g) = |\mathrm{Fix}(g)|$ (Example 5.9), and

$$\langle 1_G, \chi_{\mathbb{C}^4} \rangle_{S_4} = \frac{1}{24}(4 + 6 \cdot 2 + 3 \cdot 0 + 8 \cdot 1 + 6 \cdot 0) = 1.$$

Thus the trivial representation occurs once. As for $\mathbb{C}^3$ (§ 5.5), $\mathbb{C}^4 = \mathbb{C}(1,1,1,1) \oplus W$, where

$$W = \{x : x_1 + x_2 + x_3 + x_4 = 0\}.$$

So $\chi_W = |\mathrm{Fix}| - 1$ (Proposition 6.2(iii)), and

$$\langle \chi_W, \chi_W \rangle_{S_4} = \frac{1}{24}(9 + 6 \cdot 1 + 3 \cdot 1 + 8 \cdot 0 + 6 \cdot 1) = 1.$$

The character-norm criterion in Corollary 6.8 shows that W is irreducible.

Its sign twist. The tensor-product character formula gives $\mathrm{sgn} \cdot \chi_W$. Its squared norm is unchanged because $|\mathrm{sgn}(g)| = 1$ for every g, so $\mathrm{sgn} \otimes W$ is irreducible. The values on transpositions distinguish it from W.

The lift. § 2.6 gives a homomorphism $\pi : S_4 \to S_3$, onto, with kernel the Klein four-group V_4; by Theorem 6.15(ii), $\rho_E \circ \pi$ is irreducible. To read off its character, find the class of $\pi(\sigma)$:

- $\pi(\sigma) = e$ exactly for $\sigma \in V_4$, the class (2^2) with e;
- a transposition and a 3-cycle go to the same types,

$$\pi\big((1\,2)\big) = (2\,3), \qquad \pi\big((1\,2\,3)\big) = (1\,3\,2);$$

- $(1\,2\,3\,4) \notin V_4$, but

$$(1\,2\,3\,4)^2 = (1\,3)(2\,4) \in V_4,$$

  so $\pi((1\,2\,3\,4))$ has order 2: a transposition.

So $\chi_E \circ \pi$ is read off Example 6.4.

The table. The five rows are distinct irreducible characters, so Corollary 6.8(v) shows that they complete the table.

cycle type	(1^4)	$(2,1^2)$	(2^2)	$(3,1)$	(4)
size	1	6	3	8	6
1_G	1	1	1	1	1
sgn	1	−1	1	1	−1
χ_W	3	1	−1	0	−1
$\mathrm{sgn} \cdot \chi_W$	3	−1	−1	0	1
$\chi_E \circ \pi$	2	0	2	−1	0

For example, the rows of χ_W and $\chi_E \circ \pi$ are orthogonal:

$$\langle \chi_W, \chi_E \circ \pi \rangle_{S_4} = \frac{1}{24}(6 + 0 + 3 \cdot (-2) + 0 + 0) = 0.$$

The squares of the degrees add up to the order,

$$1 + 1 + 9 + 9 + 4 = 24 = |S_4|\,,$$

and Lecture 7 proves that $\sum_U (\dim U)^2 = |G|$ for every G.

Self-check. Find the kernel of each irreducible character of S_4. *Answer:* by Proposition 6.17(ii), read off where each row equals its degree:

χ	1_G	sgn	χ_W	$\mathrm{sgn}\cdot\chi_W$	$\chi_E \circ \pi$
$\ker\chi$	S_4	A_4	$\mathbf{1}$	$\mathbf{1}$	V_4

These recover § 2.6's normal subgroups.

Summary.

- *The main results:* $\dim \mathrm{Hom}_G(U,V) = \langle \chi_U, \chi_V \rangle_G$ (Theorem 6.7), so the irreducible characters are orthonormal and a representation is determined by its character (Corollary 6.8); the characters of constructions and lifts (Theorems 6.13 and 6.15); the values bound and the kernel (Proposition 6.17).
- *Handed on:* at most as many irreducibles as classes (Corollary 6.8(v)), to Lecture 7, which proves equality; the values bound and kernels to Lecture 9's Burnside theorem, and $\chi(g^2)$ to its indicator; the norm-one guard to Lecture 13; the table of S_4 to Lectures 8 and 13.
- *Next:* Lecture 7 asks where the irreducibles live, and why there are as many as classes.

References

Serre, *Linear Representations of Finite Groups*: characters, Props 1–3, pp. 10–11; the dual, Exercise 2.3, p. 12; orthonormality and multiplicities, Thms 3–4 and their corollaries, pp. 15–16; $|\chi(s)| \le n$, Exercise 6.7, p. 53. James and Liebeck, *Representations and Characters of Groups*: the values bound and $\ker\rho$, Thm 13.11, p. 124; $\dim \mathrm{Hom}_{\mathbb{C}G}(V,W)$, Thm 14.24, p. 145; lifting, Thm 17.3, p. 169, and linear characters, Thm 17.11, p. 174; the two squares, Prop. 19.14, p. 198; the table of S_4, §18.1, pp. 179–180.[2]

[2] James and Liebeck use right modules, $v \mapsto vg$, and write $\chi(1)$ for the degree.